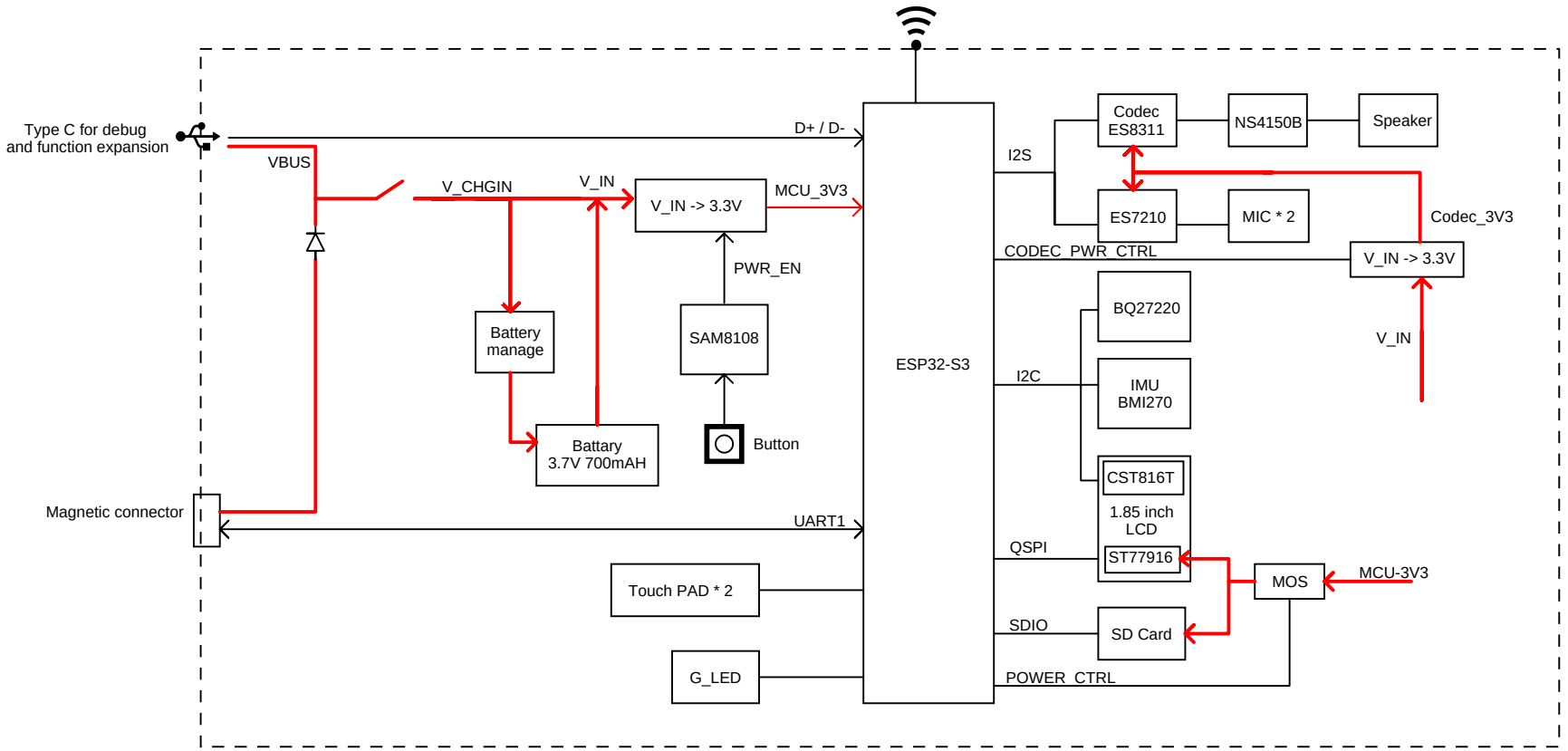
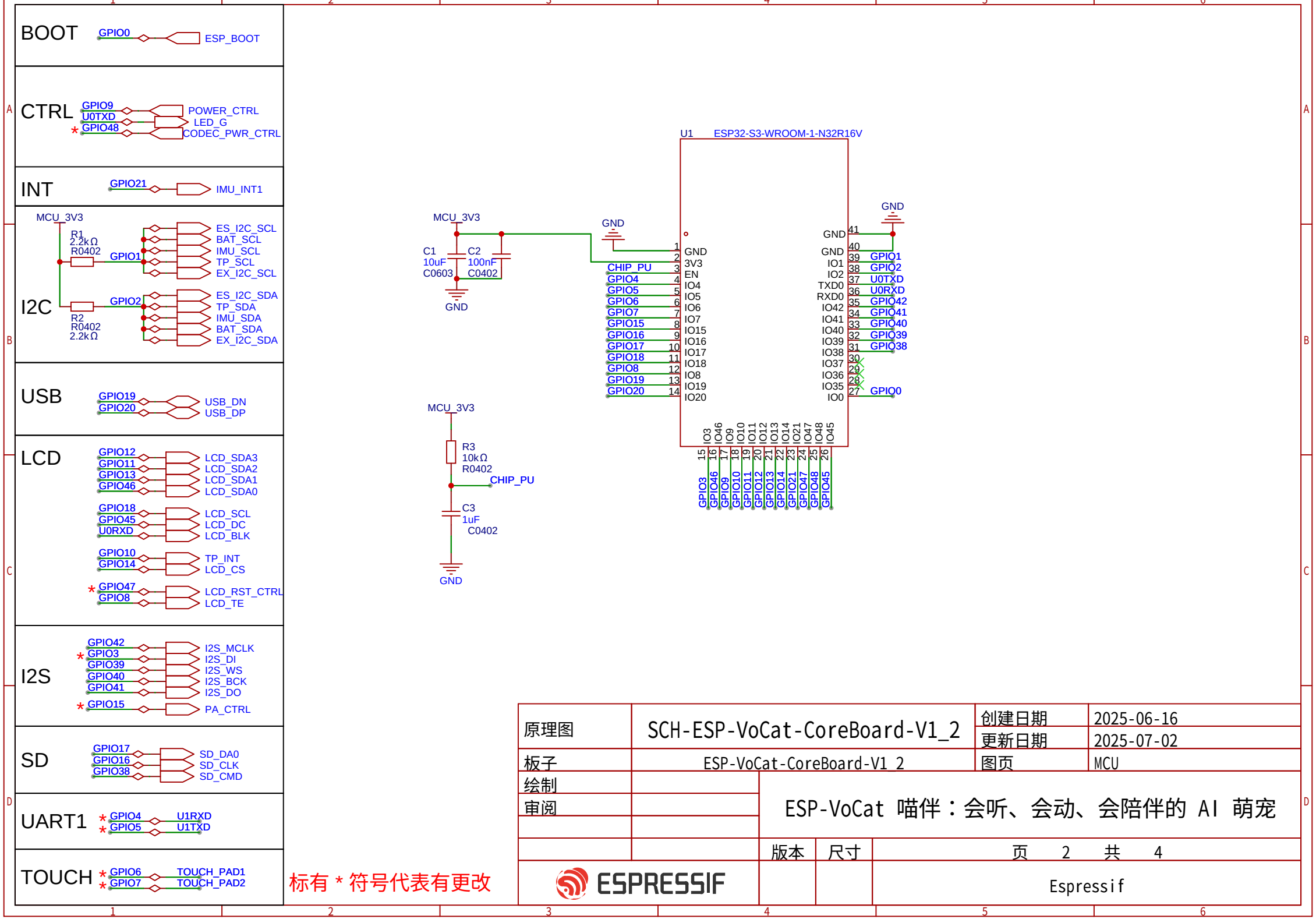
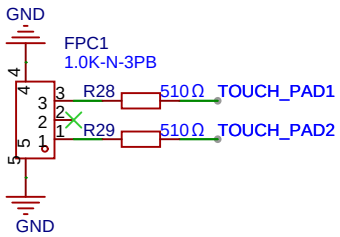


原理图	SCH-ESP-VoCat-CoreBoard-V1_2	创建日期	2025-06-16
		更新日期	2026-03-10
板子	ESP-VoCat-CoreBoard-V1_2	图页	Overview
绘制		ESP-VoCat 喵伴：会听、会动、会陪伴的 AI 萌宠	
审阅			
		版本 / 尺寸	页 1 共 4
ESPRESSIF			Espressif

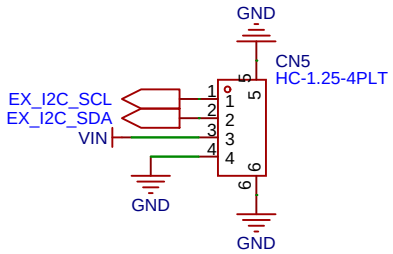

BOOT
GPIO0
ESP_BOOT
CTRL
GPIO9
U0TXD
GPIO48
POWER_CTRL
LED_G
CODEC_PWR_CTRL
INT
GPIO21
IMU_INT1
I2C
MCU_3V3
R1
2.2kΩ
R0402
GPIO1
ES_I2C_SCL
BAT_SCL
IMU_SCL
TP_SCL
EX_I2C_SCL
R2
R0402
2.2kΩ
GPIO2
ES_I2C_SDA
TP_SDA
IMU_SDA
BAT_SDA
EX_I2C_SDA
USB
GPIO19
GPIO20
USB_DN
USB_DP
LCD
GPIO12
GPIO11
GPIO13
GPIO46
LCD_SDA3
LCD_SDA2
LCD_SDA1
LCD_SDA0
GPIO18
GPIO45
U0RXD
LCD_SCL
LCD_DC
LCD_BLK
GPIO10
GPIO14
TP_INT
LCD_CS
GPIO47
GPIO8
LCD_RST_CTRL
LCD_TE
I2S
GPIO42
GPIO3
GPIO39
GPIO40
GPIO41
GPIO15
I2S_MCLK
I2S_DI
I2S_WS
I2S_BCK
I2S_DO
PA_CTRL
SD
GPIO17
GPIO16
GPIO38
SD_DA0
SD_CLK
SD_CMD
UART1
GPIO4
GPIO5
U1RXD
U1TXD
TOUCH
GPIO6
GPIO7
TOUCH_PAD1
TOUCH_PAD2
标有 * 符号代表有更改
U1
ESP32-S3-WROOM-1-N32R16V
MCU_3V3
C1
10uF
C0603
C2
100nF
C0402
GND
CHIP_PU
GPIO4
GPIO5
GPIO6
GPIO7
GPIO15
GPIO16
GPIO17
GPIO18
GPIO8
GPIO19
GPIO20
GPIO1
GPIO2
U0TXD
U0RXD
GPIO42
GPIO41
GPIO40
GPIO39
GPIO38
GPIO0
R3
10kΩ
R0402
C3
1uF
C0402
原理图
SCH-ESP-VoCat-CoreBoard-V1_2
创建日期
2025-06-16
更新日期
2025-07-02
板子
ESP-VoCat-CoreBoard-V1_2
图页
MCU
绘制
审阅
ESP-VoCat 喵伴：会听、会动、会陪伴的 AI 萌宠
版本
尺寸
页 2 共 4
ESPRESSIF
Espressif

SD

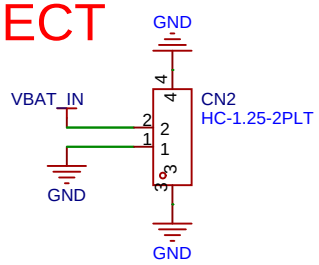
TOUCH

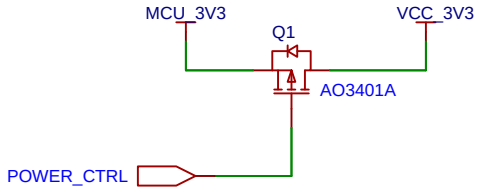


I2C EXP

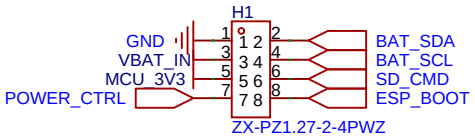


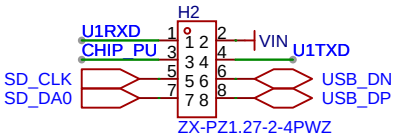
BAT CONNECT



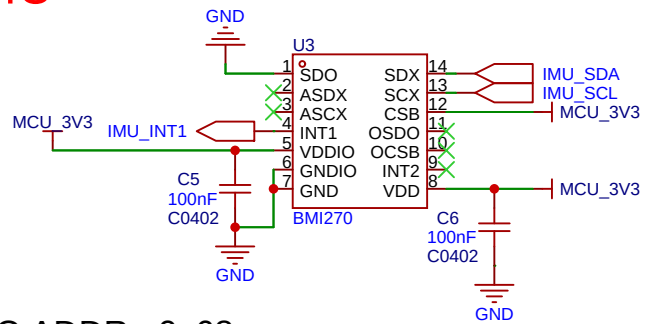


B2B CONNECT



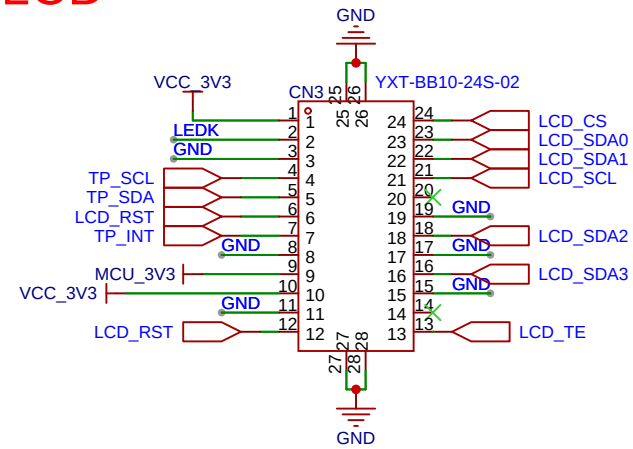


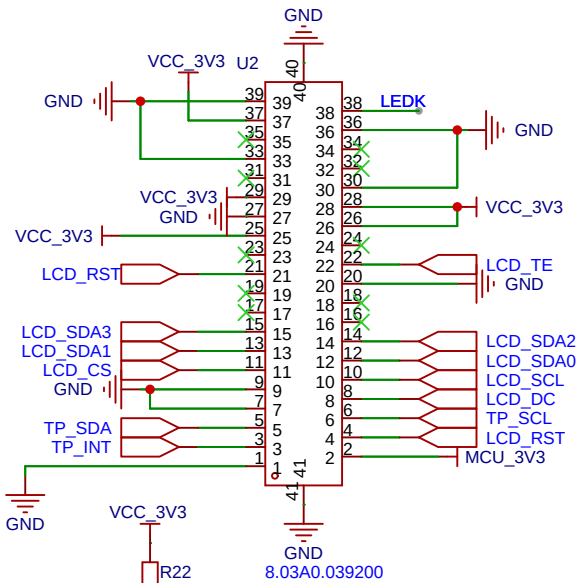
IMU

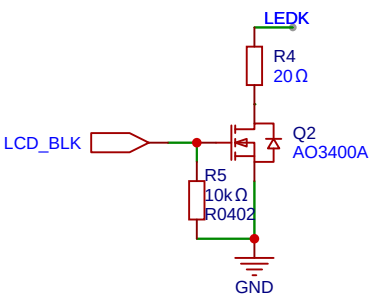


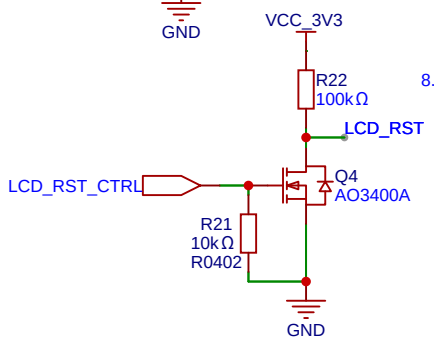
IIC ADDR : 0x68

LCD



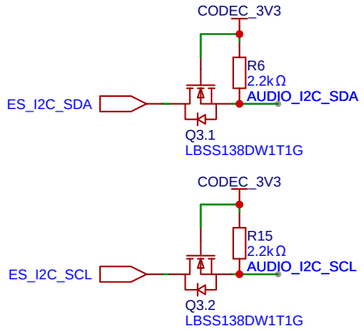




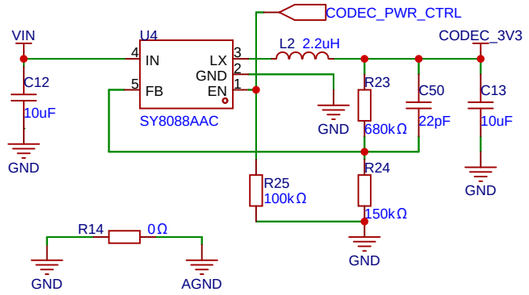


原理图	SCH-ESP-VoCat-CoreBoard-V1_2	创建日期	2025-06-16
		更新日期	2025-07-14
板子	ESP-VoCat-CoreBoard-V1_2	图页	Interface
绘制		ESP-VoCat 喵伴：会听、会动、会陪伴的 AI 萌宠	
审阅			
		版本 / 尺寸	页 3 共 4
ESPRESSIF			Espressif

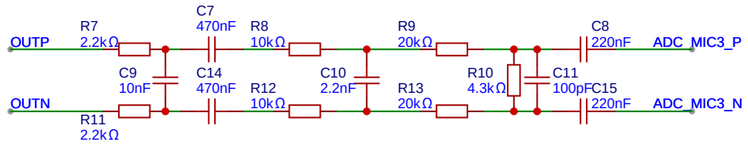
CODEC_IIC



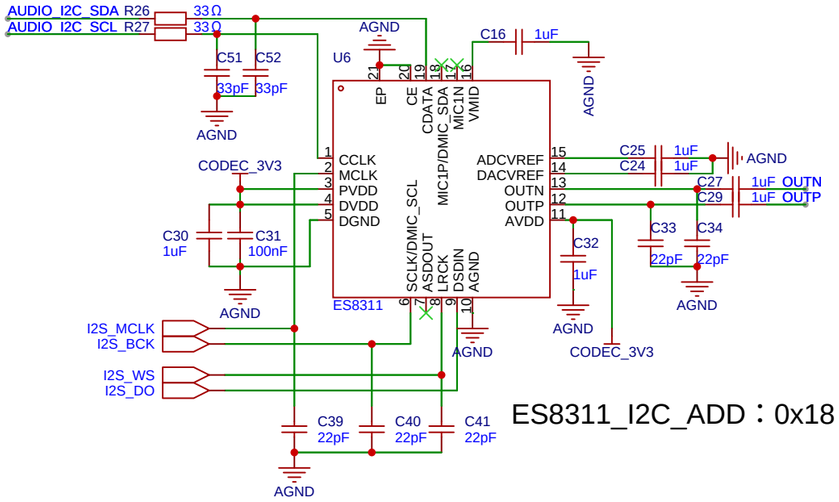
CODEC 3V3 DCDC



AEC

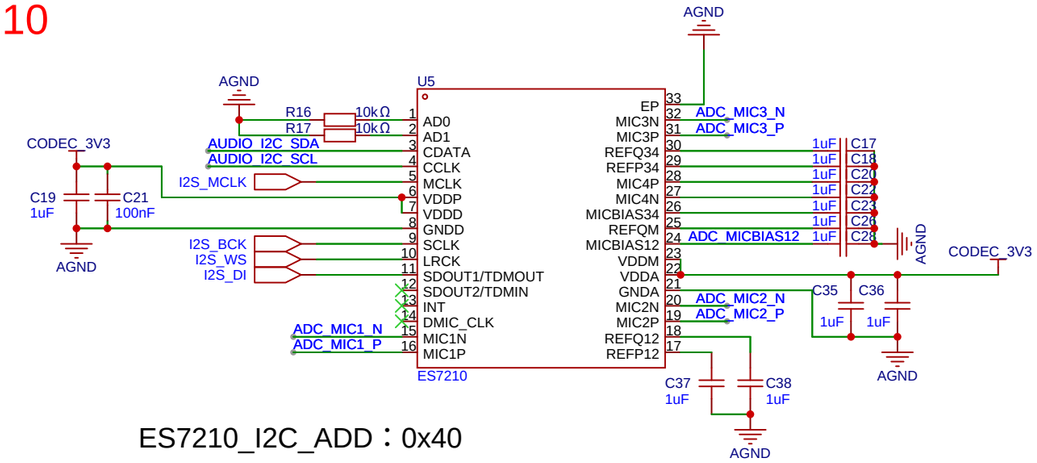


ES8311



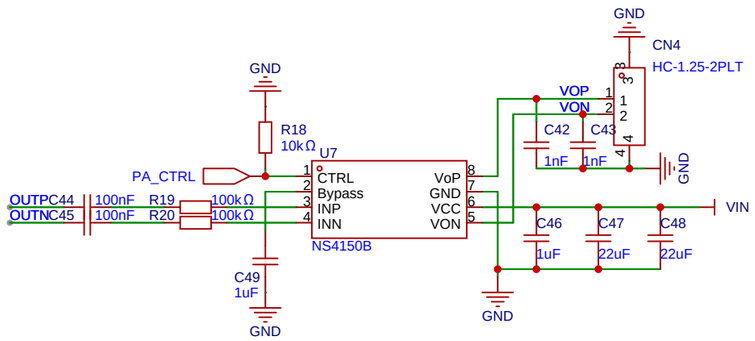
ES8311_I2C_ADD：0x18

ES7210



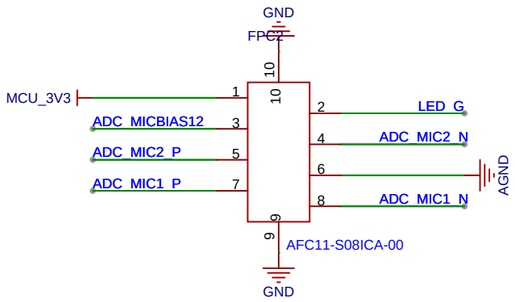
ES7210_I2C_ADD：0x40

PA



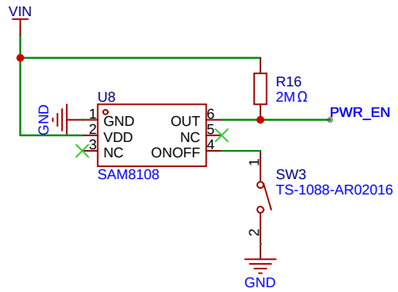
放大倍数 = 240k / RIN

MIC CONNECT

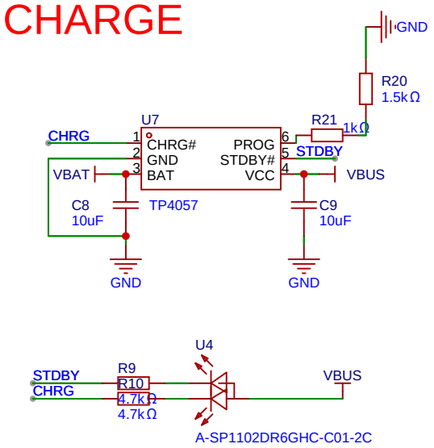


原理图	SCH-ESP-VoCat-CoreBoard-V1_2	创建日期	2025-06-16
		更新日期	2025-06-23
板子	ESP-VoCat-CoreBoard-V1_2	图页	codec
绘制		ESP-VoCat 喵伴：会听、会动、会陪伴的 AI 萌宠	
审阅			
		版本	尺寸
ESPRESSIF		页 4 共 4	
		Espressif	

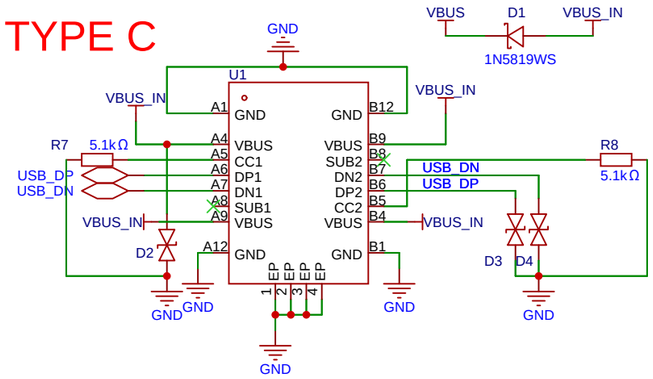
POWER ON/OFF MANAGEMENT



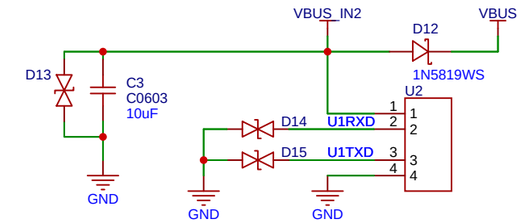
BAT CHARGE



TYPE C



Magnetic connector

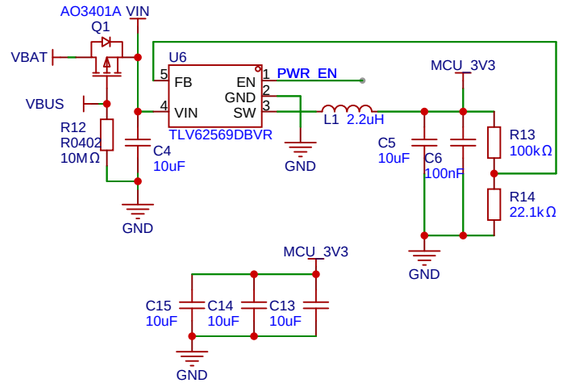


Soldered screw hole

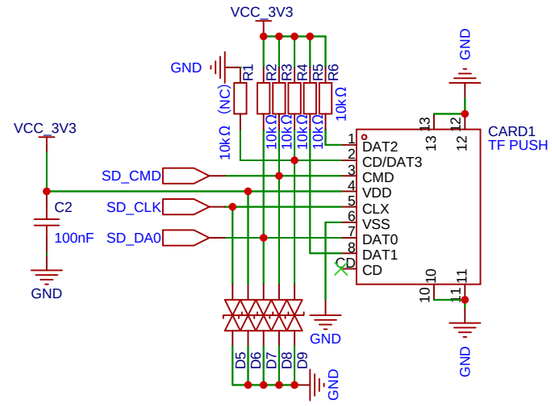


DCDC

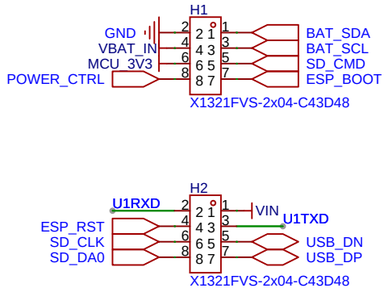
Vout = 0.6 x (1 + R10/R12) = 3.315V



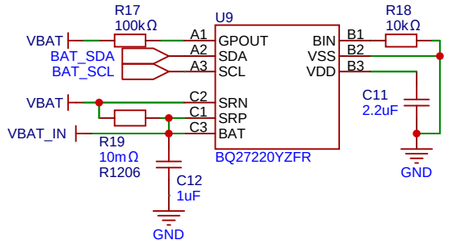
SD CARD



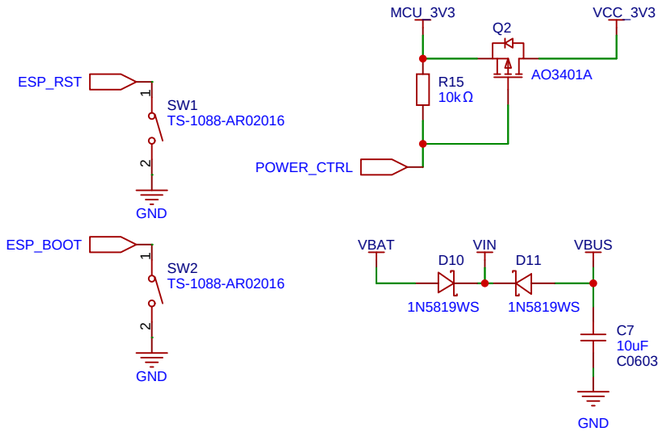
B2B connector



BAT MANAGEMENT



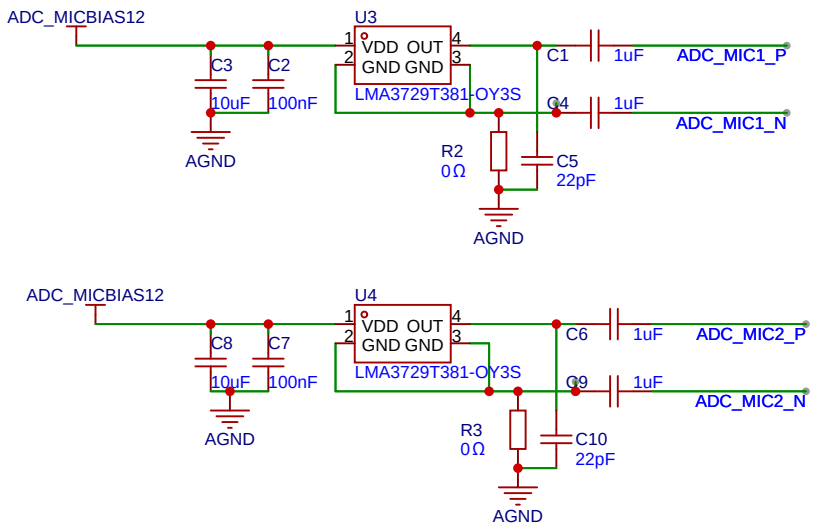
ESP_RST / ESP_BOOT / POWER_CTRL



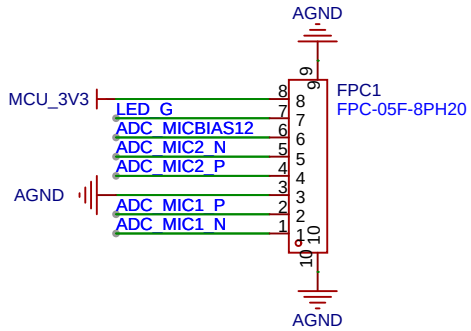
原理图	SCH-ESP-VoCat-BaseBoard-V1_2	创建日期	2025-06-25
		更新日期	2026-03-10
板子	ESP-VoCat-BaseBoard-V1_2	图页	P1
绘制		ESP-VoCat 喵伴：会听、会动、会陪伴的 AI 萌宠	
审阅			
		版本 / 尺寸	页 1 共 1
ESPRESSIF		Espressif	

TYPE-C 16PIN 2MD(073)

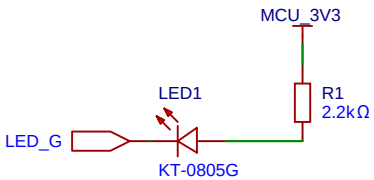
MIC



FPC CONNECT



LED



原理图	SCH-ESP-VoCat-MicBoard-V1_2	创建日期	2025-06-16
		更新日期	2025-07-14
板子	ESP-VoCat-MicBoard-V1_2	图页	P1
绘制		ESP-VoCat 喵伴：会听、会动、会陪伴的 AI 萌宠	
审阅			
		版本 / 尺寸	页 1 共 1
ESPRESSIF			Espressif